

Statistical and Econometric Methods for Transportation Data Analysis

Chapter 16 and 17 – Random Parameter and Latent Class Models

Example 16.3 (Extended)

Latent Class and Correlated Random Parameters Count-Data Models

You are given accident, environmental, traffic, and roadway geometric data from 275 segments of highway in Washington State. The data are from 1990. Your task is to estimate a count-data model for the total number of accidents on these segments.

Using data for Example 16.3 in the text, in this example you will estimate:

1. A latent class negative binomial model (see Chapter 17)
2. A latent class negative binomial model with logit class probability functions (see Chapter 17).
3. A standard random parameters negative binomial model with un-correlated random parameters
4. A correlated random parameters negative binomial model (see Sections 16.10, 16.11, 16.12 and 16.13 in the text)

In your write-up include:

1. The results of your best model specification.
2. A discussion of the logical process that led you to the selection of your final specification (the theory behind the inclusion of your selected variables). Include *t*-statistics and justify the signs of your variables.

Variables available for your specification are (in file Ex16-3.txt):

Variable Number	Explanation
ID	Segment ID number
FREQ	Number of accidents
ROUTE	Route Number
LENGTH	Segment length in miles
INCLANES	Number of lanes in increasing milepost direction
DECLANES	Number of lanes in decreasing milepost direction
WIDTH	Total combined width of all lanes
MIMEDSH	Minimum median shoulder in feet
MXMEDSH	Maximum median shoulder in feet
SPEED	Speed limit (mi/h)
URB	Indicates urban area (1=yes, 0=no)
FC	Functional class (1=local, 2=collector, 3=arterial, 4=principal arterial, 5=interstate)
AADT	Average Annual Daily Traffic
SINGLE	Daily percentage of single unit trucks
DOUBLE	Daily percentage of tractor and trailer trucks
TRAIN	Daily percentage of tractor and two-trailer trucks
PEAKHR	Percent of daily traffic in the peak hour
GRADEBR	Number of grade breaks in the segment
MIGRADE	Minimum grade in the segment
MXGRADE	Maximum grade in the segment
MXGRDIFF	Maximum grade difference in the segment
TANGENT	Tangent length in the segment
CURVES	Number of curves in the segment

MINRAD	Minimum radius in feet
ACCESS	Segment access control (0=none, 1=partial, 3=full)
MEDWIDTH	Median width (1=less than 30ft; 2=30 to 40ft; 3=40 to 50ft; 4=50 to 60ft to 5=high)
FRICTION	Friction value (0 to 100 with 100 being high)
ADTLANE	Average daily travel per lane
SLOPE	Segment slope (0=flat, 1=slight, 2=medium, 3=high)
INTECHAG	Indicates number of interchanges in the segment
AVEPRE	Average precipitation per month in inches
AVESNOW	Average snowfall per month in inches

```
| -> read;nvar=32;nobs=275;names=ID,FREQ,ROUTE,LENGTH,INCLANES,DECLANES,WIDTH,
MIMEDSH,MXMEDSH,SPEED,URB,FC,AADT,SINGLE,DOUBLE,TRAIN,PEAKHR,GRADEBR,
MIGRADE,MXGRADE,MXGRDIFF,TANGENT,CURVES,MINRAD,ACCESS,MEDWIDTH,
FRICTION,ADTLANE,SLOPE,INTECHAG,AVEPRE,AVESNOW
;FILE=C:\Users\Fred\Box\UDrive Data\flm\00Work-Purdue\Book\Book2-3e-Data\Ex16-
```

3.txt\$

```
| -> create;laneadt=aadt/(inclanes+declanes)$
| -> create;lnlanadt=log(laneadt)$
| -> create;lnaadt=log(aadt)$
| -> create;density=laneadt/length$
| -> create;if(friction<=30)lowfri=1$
| -> create;if(friction>30&friction<50)medfri=1$
| -> create;if(friction>=50)hifri=1$
| -> create;curvmile=curves/length$
| -> create;if(curvmile<=0.5)lowcvmil=1;(else)lowcvmil=0$
| -> create;if(curvmile>0.5&curvmile<=2.5)medcvmil=1;(else)medcvmil=0$
| -> create;if(curvmile>2.5)hiccvmil=1;(else)hiccvmil=0$
| -> create;truck=single+double+train$
| -> create;notruck=truck*aadt$
| -> create;if(medwidth=1)med030=1$
| -> create;if(medwidth=2)med3040=1$
| -> create;if(medwidth=3)med4050=1$
| -> create;if(medwidth=4)med5060=1$
| -> create;if(medwidth=5)med60=1$
| -> create;if(speed<=50)speed1=1$
| -> create;if(speed<=55)speed2=1$
| -> create;if(speed>55)speed3=1$
| -> create;if(speed>=55)speed4=1$
| -> create;if(fc=1)local=1$
| -> create;if(fc=5)intstate=1$
| -> create;if(access=0)none =1$
| -> create;if(access=1)partial=1$
| -> create;if(access=2)full =1$
| -> create;if(slope=0)flat=1$
| -> create;if(slope=1)slight=1$
| -> create;if(slope=2)medium=1$
| -> create;if(slope=0 |slope= 1)slpflat=1;(else)slpflat=0$
| -> create;if(slope=2)slpmed=1;(else)slpmed=0$
```

```

|-> create;if(avepre<=1.5)lowpre=1;(else)lowpre=0$
|-> create;if(avepre>1.5&avepre<=2.5)medpre=1;(else)medpre=0$
|-> create;if(avepre>2.5)hipre=1;(else)hipre=0$
|-> create;if(avesnow<=1)norsnow=1$
|-> create;if(avesnow>1)hisnow=1$
|-> create;lanewid=(inclanes+declanes)/width$
|-> dstat;rhs=lanewid$

```

Variable	Mean	Standard Deviation	Minimum	Maximum	Cases	Missing Values
LANEWID	.080906	.006445	.039216	.086957	275	0

```

|-> create;if(lanewid<12)nlanwid=1;(else)nlanwid=0$
|-> create;if(lanewid>12)wlanwid=1;(else)wlanwid=0$
|-> create;expose=aadt*length*365/1000000000$
|-> create;cpm=curves/length$
|-> create;intpm=intechag/length$
create;gbrpm=gradebr/length$

```

Latent -Class

```

negbin;lhs=freq;rhs=one,expose,lowpre,gbrpm,intpm,cpm,hisnow,friction
;lcm;pts=2$

```

```

-----
Poisson Regression Start Values for FREQ
Dependent variable          FREQ
Log likelihood function      -1930.44685
Restricted log likelihood    -3178.57212
Chi squared [ 7](P= .000)    2496.25054
Significance level           .000000
McFadden Pseudo R-squared    .3926685
Estimation based on N =     275, K = 8
Inf.Cr.AIC = 3876.9 AIC/N = 14.098

```

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
Constant	3.47694***	.16673	20.85	.0000	3.15015	3.80373
EXPOSE	1.04888***	.02179	48.13	.0000	1.00617	1.09159
LOWPRE	-.45972***	.04068	-11.30	.0000	-.53946	-.37999
GBRPM	.00984	.01049	.94	.3482	-.01072	.03039
INTPM	.26642***	.02607	10.22	.0000	.21533	.31751
CPM	-.21030***	.01887	-11.15	.0000	-.24728	-.17332
HISNOW	-.16301***	.03197	-5.10	.0000	-.22567	-.10034
FRICITION	-.01499***	.00329	-4.55	.0000	-.02144	-.00854

```

***, **, * ==> Significance at 1%, 5%, 10% level.
Model was estimated on Mar 30, 2020 at 04:38:11 PM

```

```

-----
Iterative procedure has converged
Normal exit: 50 iterations. Status=0, F= .9401885D+03

```

```

-----
Latent Class / Panel NegBnReg Model
Dependent variable          FREQ
Log likelihood function      -940.18855
Restricted log likelihood    -1930.44685
Chi squared [ 11](P= .000)   1980.51662
Significance level           .000000
McFadden Pseudo R-squared    .5129684
Estimation based on N =     275, K = 19

```

Inf.Cr.AIC = 1918.4 AIC/N = 6.976
 Sample is 1 pds and 275 individuals
 Latent class model with 2 latent classes
 Negative binomial regression model

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
Model parameters for latent class 1.....						
Constant	1.77652***	.60789	2.92	.0035	.58509	2.96795
EXPOSE	2.82737***	.26255	10.77	.0000	2.31278	3.34196
LOWPRE	-.33465***	.12103	-2.76	.0057	-.57187	-.09743
GBRPM	.08621***	.02124	4.06	.0000	.04458	.12783
INTPM	.33500***	.07826	4.28	.0000	.18161	.48838
CPM	-.08092	.05548	-1.46	.1447	-.18966	.02781
HISNOW	.06144	.09874	.62	.5338	-.13209	.25496
FRICITION	.00125	.01035	.12	.9039	-.01904	.02154
Alpha	4.34981***	.77034	5.65	.0000	2.83997	5.85965
Model parameters for latent class 2.....						
Constant	6.79519	4.28087	1.59	.1124	-1.59516	15.18555
EXPOSE	1.34809**	.54266	2.48	.0130	.28449	2.41168
LOWPRE	-.81472	.93330	-.87	.3827	-2.64396	1.01452
GBRPM	-.18550	.21522	-.86	.3887	-.60732	.23633
INTPM	-.55301	.50459	-1.10	.2731	-1.54198	.43596
CPM	-.49793	.33931	-1.47	.1422	-1.16296	.16710
HISNOW	-.81487	.60321	-1.35	.1767	-1.99713	.36740
FRICITION	-.07137	.07647	-.93	.3507	-.22124	.07851
Alpha	1.05338**	.40983	2.57	.0102	.25012	1.85663
Estimated prior probabilities for class membership.....						
Class1Pr	.76588***	.06305	12.15	.0000	.64230	.88947
Class2Pr	.23412***	.06305	3.71	.0002	.11053	.35770

***, **, * ==> Significance at 1%, 5%, 10% level.
 Model was estimated on Mar 30, 2020 at 04:38:12 PM

Latent-Class with Class Probability Functions

```
| -> negbin; lhs=freq; rhs=one,expose,lowpre,gbrpm,intpm,cpm,hisnow,friction
; lcm=one,expose; pts=2$
```

Poisson Regression Start Values for FREQ
 Dependent variable FREQ
 Log likelihood function -1930.44685
 Restricted log likelihood -3178.57212
 Chi squared [7](P= .000) 2496.25054
 Significance level .00000
 McFadden Pseudo R-squared .3926685
 Estimation based on N = 275, K = 8
 Inf.Cr.AIC = 3876.9 AIC/N = 14.098

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
Constant	3.47694***	.16673	20.85	.0000	3.15015	3.80373
EXPOSE	1.04888***	.02179	48.13	.0000	1.00617	1.09159
LOWPRE	-.45972***	.04068	-11.30	.0000	-.53946	-.37999
GBRPM	.00984	.01049	.94	.3482	-.01072	.03039
INTPM	.26642***	.02607	10.22	.0000	.21533	.31751
CPM	-.21030***	.01887	-11.15	.0000	-.24728	-.17332
HISNOW	-.16301***	.03197	-5.10	.0000	-.22567	-.10034
FRICITION	-.01499***	.00329	-4.55	.0000	-.02144	-.00854

***, **, * ==> Significance at 1%, 5%, 10% level.

Model was estimated on Mar 30, 2020 at 04:59:51 PM

Iterative procedure has converged

Normal exit: 59 iterations. Status=0, F= .9338371D+03

Latent Class / Panel NegBnReg Model

Dependent variable FREQ
 Log likelihood function -933.83707
 Restricted log likelihood -1930.44685
 Chi squared [13](P= .000) 1993.21956
 Significance level .000000
 McFadden Pseudo R-squared .5162586
 Estimation based on N = 275, K = 20
 Inf.Cr.AIC = 1907.7 AIC/N = 6.937
 Sample is 1 pds and 275 individuals
 Latent class model with 2 latent classes
 Negative binomial regression model

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
Model parameters for latent class 1.....						
Constant	1.03907*	.62167	1.67	.0946	-.17937	2.25752
EXPOSE	6.90031***	.87138	7.92	.0000	5.19243	8.60818
LOWPRE	-.18132	.14395	-1.26	.2078	-.46347	.10082
GBRPM	.13900***	.02496	5.57	.0000	.09007	.18792
INTPM	.24185***	.07580	3.19	.0014	.09328	.39042
CPM	-.06283	.05522	-1.14	.2552	-.17105	.04540
HISNOW	.14752	.11399	1.29	.1956	-.07589	.37093
FRICITION	-.00130	.01152	-.11	.9100	-.02388	.02127
Alpha	4.29362***	.97330	4.41	.0000	2.38600	6.20125
Model parameters for latent class 2.....						
Constant	3.89765**	1.83140	2.13	.0333	.30817	7.48713
EXPOSE	1.43017***	.36545	3.91	.0001	.71389	2.14644
LOWPRE	-.08827	.56938	-.16	.8768	-1.20422	1.02769
GBRPM	-.22511	.15946	-1.41	.1580	-.53765	.08743
INTPM	-.09078	.35511	-.26	.7982	-.78677	.60522
CPM	-.11575	.24095	-.48	.6309	-.58801	.35650
HISNOW	-.21533	.33224	-.65	.5169	-.86651	.43584
FRICITION	-.01622	.03585	-.45	.6510	-.08650	.05405
Alpha	1.31509***	.19106	6.88	.0000	.94062	1.68956
Estimated prior probabilities for class membership.....						
ONE_1	3.84194***	1.07573	3.57	.0004	1.73355	5.95033
EXPOSE_1	-13.2268***	4.64305	-2.85	.0044	-22.3270	-4.1266
ONE_2	0.0	(Fixed Parameter).....				
EXPOSE_2	0.0	(Fixed Parameter).....				

***, **, * ==> Significance at 1%, 5%, 10% level.

Fixed parameter ... is constrained to equal the value or had a nonpositive st.error because of an earlier problem.

Model was estimated on Mar 30, 2020 at 04:59:51 PM

Prior class probabilities at data means for LCM variables

Class 1	Class 2
.60624	.39376

Un-Correlated Random Parameteres

```
| -> negbin; lhs=freq; rhs=one, expose, lowpre, gbrpm, intpm, cpm, hisnow, friction
;rpm;pts=200;halton
;fcu=expose(n),intpm(n),cpm(n);marginal effects$
```

Poisson Regression Start Values for FREQ

```
Dependent variable      FREQ
Log likelihood function  -1930.44685
Restricted log likelihood -3178.57212
Chi squared [ 7](P= .000) 2496.25054
Significance level      .00000
McFadden Pseudo R-squared .3926685
Estimation based on N = 275, K = 8
Inf.Cr.AIC = 3876.9 AIC/N = 14.098
```

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
Constant	3.47694***	.16673	20.85	.0000	3.15015	3.80373
LOWPRE	-.45972***	.04068	-11.30	.0000	-.53946	-.37999
GBRPM	.00984	.01049	.94	.3482	-.01072	.03039
HISNOW	-.16301***	.03197	-5.10	.0000	-.22567	-.10034
FRICITION	-.01499***	.00329	-4.55	.0000	-.02144	-.00854
EXPOSE	1.04888***	.02179	48.13	.0000	1.00617	1.09159
INTPM	.26642***	.02607	10.22	.0000	.21533	.31751
CPM	-.21030***	.01887	-11.15	.0000	-.24728	-.17332

***, **, * ==> Significance at 1%, 5%, 10% level.

Model was estimated on Apr 01, 2020 at 07:57:56 PM

Iterative procedure has converged

Normal exit: 29 iterations. Status=0, F= .9626575D+03

Random Coefficients NegBnReg Model

```
Dependent variable      FREQ
Log likelihood function  -962.65751
Restricted log likelihood -1930.44685
Chi squared [ 3](P= .000) 1935.57870
Significance level      .00000
McFadden Pseudo R-squared .5013292
Estimation based on N = 275, K = 12
Inf.Cr.AIC = 1949.3 AIC/N = 7.088
Sample is 1 pds and 275 individuals
Simulation based on 200 Halton draws
Negative binomial regression model
```

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	

	Nonrandom parameters.....					
Constant	2.88190***	.49031	5.88	.0000	1.92091	3.84288
LOWPRE	-.41734***	.11059	-3.77	.0002	-.63410	-.20058
GBRPM	-.05080*	.02780	-1.83	.0676	-.10529	.00369
HISNOW	-.05048	.09075	-.56	.5780	-.22836	.12739
FRICITION	-.01281	.00907	-1.41	.1580	-.03058	.00497
	Means for random parameters.....					
EXPOSE	2.50802***	.17314	14.49	.0000	2.16867	2.84736
INTPM	.16909**	.07432	2.28	.0229	.02343	.31475
CPM	-.17775***	.04913	-3.62	.0003	-.27405	-.08145

	Scale parameters for dists. of random parameters.....					
EXPOSE	.83830***	.10119	8.28	.0000	.63997	1.03662
INTPM	.51164***	.05508	9.29	.0000	.40369	.61959
CPM	.03133	.02475	1.27	.2057	-.01719	.07984
	Dispersion parameter for NegBin distribution.....					
ScalParm	2.96187***	.28685	10.33	.0000	2.39965	3.52408

***, **, * ==> Significance at 1%, 5%, 10% level.

Model was estimated on Apr 01, 2020 at 07:58:28 PM

Partial derivatives of expected val. with respect to the vector of characteristics.

They are computed at the means of the Xs.

Conditional Mean at Sample Point .0000

Scale Factor for Marginal Effects .0000

FREQ	Partial Effect	Elasticity	z	Prob. z >Z*	95% Confidence Interval	
LOWPRE	-4.90070***	-.14417	-3.77	.0002	-7.44958	-2.35183
GBRPM	-.59656*	-.09585	-1.79	.0739	-1.25074	.05763
HISNOW	-.59280	-.01946	-.55	.5789	-2.68628	1.50069
FRICITION	-.15037	-.63582	-1.42	.1545	-.35736	.05663
EXPOSE	29.4509***	.64667	9.35	.0000	23.2782	35.6235
INTPM	1.98553**	.08594	2.27	.0234	.26855	3.70252
CPM	-2.08725***	-.25597	-4.86	.0000	-2.92862	-1.24588

z, prob values and confidence intervals are given for the partial effect

***, **, * ==> Significance at 1%, 5%, 10% level.

Model was estimated on Apr 01, 2020 at 07:58:28 PM

Correlated Random Parameteres

```
| -> negbin; lhs=freq; rhs=one, expose, lowpre, gbrpm, intpm, cpm, hisnow, friction
;rpm;pts=200;halton
;fcu=expose(n),intpm(n),cpm(n);marginal effects;correlated$
```

Poisson Regression Start Values for FREQ

Dependent variable FREQ

Log likelihood function -1930.44685

Restricted log likelihood -3178.57212

Chi squared [7](P= .000) 2496.25054

Significance level .00000

McFadden Pseudo R-squared .3926685

Estimation based on N = 275, K = 8

Inf.Cr.AIC = 3876.9 AIC/N = 14.098

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
Constant	3.47694***	.16673	20.85	.0000	3.15015	3.80373
LOWPRE	-.45972***	.04068	-11.30	.0000	-.53946	-.37999
GBRPM	.00984	.01049	.94	.3482	-.01072	.03039
HISNOW	-.16301***	.03197	-5.10	.0000	-.22567	-.10034
FRICITION	-.01499***	.00329	-4.55	.0000	-.02144	-.00854
EXPOSE	1.04888***	.02179	48.13	.0000	1.00617	1.09159
INTPM	.26642***	.02607	10.22	.0000	.21533	.31751
CPM	-.21030***	.01887	-11.15	.0000	-.24728	-.17332

***, **, * ==> Significance at 1%, 5%, 10% level.

Model was estimated on Apr 01, 2020 at 08:00:41 PM

Iterative procedure has converged
Normal exit: 32 iterations. Status=0, F= .9616611D+03

Random Coefficients NegBnReg Model
Dependent variable FREQ
Log likelihood function -961.66107
Restricted log likelihood -1930.44685
Chi squared [6](P= .000) 1937.57157
Significance level .000000
McFadden Pseudo R-squared .5018454
Estimation based on N = 275, K = 15
Inf.Cr.AIC = 1953.3 AIC/N = 7.103
Sample is 1 pds and 275 individuals
Simulation based on 200 Halton draws
Negative binomial regression model

FREQ	Coefficient	Standard Error	z	Prob. z >Z*	95% Confidence Interval	
	Nonrandom parameters.....					
Constant	2.83940***	.55136	5.15	.0000	1.75875	3.92005
LOWPRE	-.41203***	.12261	-3.36	.0008	-.65234	-.17173
GBRPM	-.05325*	.03088	-1.72	.0846	-.11377	.00727
HISNOW	-.04628	.10360	-.45	.6550	-.24934	.15677
FRICITION	-.01188	.01002	-1.19	.2357	-.03151	.00775
	Means for random parameters.....					
EXPOSE	2.47394***	.19884	12.44	.0000	2.08422	2.86365
INTPM	.21292**	.08499	2.51	.0122	.04633	.37950
CPM	-.16460***	.05602	-2.94	.0033	-.27440	-.05480
	Diagonal elements of Cholesky matrix.....					
EXPOSE	.88569***	.13230	6.69	.0000	.62638	1.14500
INTPM	.13831*	.07503	1.84	.0653	-.00874	.28536
CPM	.00689	.02699	.26	.7986	-.04601	.05978
	Below diagonal elements of Cholesky matrix C in CDC'.....					
lINT_EXP	-.53744***	.07940	-6.77	.0000	-.69306	-.38183
lCPM_EXP	-.00508	.03886	-.13	.8960	-.08124	.07108
lCPM_INT	.01571	.03597	.44	.6624	-.05479	.08620
	Dispersion parameter for NegBin distribution.....					
ScalParm	2.38978***	.19692	12.14	.0000	2.00382	2.77574

***, **, * ==> Significance at 1%, 5%, 10% level.
Model was estimated on Apr 01, 2020 at 08:01:17 PM

Implied covariance matrix of random parameters
Covariance matrix

Var_Beta	EXPOSE	INTPM	CPM
EXPOSE	.7845		
INTPM	-.4760	.3080	
CPM	-.4497E-02	.4901E-02	.3199E-03

Implied standard deviations of random parameters

Equation	Std.Dev.Alpha
1	.885693
2	.554955
3	.0178849

Implied correlation matrix of random parameters

Cor.Mat.	EXPOSE	INTPM	CPM
EXPOSE	1.00000	-.96844	-.28388
INTPM	-.96844	1.00000	.49378
CPM	-.28388	.49378	1.00000

 Partial derivatives of expected val. with
 respect to the vector of characteristics.
 They are computed at the means of the Xs.
 Conditional Mean at Sample Point .0000
 Scale Factor for Marginal Effects .0000

FREQ	Partial Effect	Elasticity	z	Prob. z >Z*	95% Confidence Interval	
LOWPRE	-5.00993***	-.14234	-3.32	.0009	-7.96783	-2.05203
GBRPM	-.64746*	-.10047	-1.68	.0928	-1.40255	.10763
HISNOW	-.56277	-.01784	-.45	.6556	-3.03585	1.91031
FRICTION	-.14442	-.58975	-1.20	.2301	-.38027	.09144
EXPOSE	30.0807***	.63788	8.12	.0000	22.8243	37.3371
INTPM	2.58884**	.10821	2.49	.0128	.54955	4.62813
CPM	-2.00133***	-.23703	-3.85	.0001	-3.01995	-.98271

 z, prob values and confidence intervals are given for the partial effect
 ***, **, * ==> Significance at 1%, 5%, 10% level.
 Model was estimated on Apr 01, 2020 at 08:01:17 PM
